Noga Aharony

Computational Biology · Systems Biology · Columbia University

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RESEARCH SYSTEMS

Evaluate
Model behavior
What do genome language models learn, use, and fail to capture?

Transfer
Microbial genomics
When do representations remain useful under realistic distribution shifts?

Detect
Biosurveillance
How can sequence models support detection across evolutionary distance?

01 Education

Doctor of Philosophy, Systems Biology · Columbia University — Current
Thesis: Systematic Evaluation Frameworks for Foundational Genome Language Models · Advisor: Mohammed AlQuraishi, PhD

Master of Science, Biology · Technion — Israel Institute of Technology — 2021
Thesis: Rapid Gene Content Alteration in Recurring Infections · Advisor: Roy Kishony, PhD

Bachelor of Science, Honours Neuroscience · McGill University — 2019
First Class Honours · Dean's Honours List · Dean's Multidisciplinary Research List

02 Current fellowships

Pivotal Research Frontier Biodefense Fellow — Aug–Oct 2026
Developing reference-free methods to detect engineered DNA in metagenomic sequencing data.

ML Alignment & Theory Scholars (MATS), Autumn 2026 — Sep–Dec 2026
Biosecurity track, mentored by Gary Abel at Fourth Eon Bio; technical research including function-based DNA synthesis screening.

03 Research experience

Mohammed AlQuraishi Lab · Columbia University · Program of Mathematical Genomics — 2022–Current
Developing evaluation frameworks and applications for genome language models across genomic context use, pathway-level reasoning, wastewater pathogen monitoring, and protein–protein interaction prediction. Co-developing GeneCtx and WAVEBench to test model behavior under realistic biological distribution shifts.

Tal Korem Lab · Columbia University · Program of Mathematical Genomics — 2021–2023
Created a model to detect small RNAs in bacterial genomes.

Itsik Pe'er Lab · Columbia University · Department of Computer Science — 2022
Developed a neural embedding model in hyperbolic spaces to improve small-protein prediction.

Roy Kishony Lab · Technion — Israel Institute of Technology · Department of Biology — 2019–2021
Built a computational pipeline to identify mobile genetic elements in recurring infections and designed experiments on eco-evolutionary dynamics in microbial communities.

Amine Kamen Lab · McGill University · Department of Bioengineering — 2018–2019
Designed and manufactured adeno-associated viruses carrying Cas9 for improved CAR-T cell engineering.

Center for Health Security · Johns Hopkins Bloomberg School of Public Health — 2018
Surveyed DNA-synthesis providers' sequence- and customer-screening practices, informing Strengthening Security for Gene Synthesis; studied safety, innovation, and community norms in DIY biology.

03 Research experience · continued

Jonathan Kimmelman Lab · McGill University · Biomedical Ethics Unit 2017–2018
Applied machine learning to analyze clinical-trial design factors that influence outcome forecasts; contributed to a meta-analysis.

Edward Ruthazer Lab · McGill University · Montreal Neurological Institute 2017
Electroporated and imaged tadpoles to study structural plasticity in the developing visual system.

Rafael Najmanovich Lab · Université de Montréal · Pharmacology and Physiology 2016–2017
Developed protein-interface software based on surrounding force fields and modeled C. difficile germination protease to identify candidate inhibitory ligands.

04 Publications and manuscripts

P01 Le Lievre T*, **Aharony N***, Wang S, Adams E, AlQuraishi M. *GeneCtx: An Evaluation Framework for Dissecting Genomic Context Use in Genome Language Models.* Submitted to NeurIPS 2026 Evaluations & Datasets track. 2026

P02 Liang C, Wagstaff J, **Aharony N**, Schmit V, Manheim D. *Managing the Transition to Widespread Metagenomic Monitoring: Policy Considerations for Future Biosurveillance.* Health Security, 21(1), 34–45. 2023

P03 Myers A, Utpala S, Talbar S, Sanborn S, Shewmake C, Donnat C, Mathe J, Lupo U, Sonthalia R, Cui X, Szwagier T, Pignet A, Bergsson A, Hauberg S, Nielson D, Sommer S, Klindt D, Hermansen E, Vaupel M, Dunn B, Xiong J, **Aharony N**, Pe'er I, Ambellan F, Hanik M, Nava-Yazdani E, von Tycowicz C, Miolane N. *ICLR 2022 Challenge for Computational Geometry & Topology: Design and Results.* arXiv. 2022

P04 Milman O*, Yelin I*, **Aharony N**, Katz R, Herzel E, Ben-Tov A, Kuint J, Gazit S, Chodick G, Patalon T, Kishony R. *Community-level evidence for SARS-CoV-2 vaccine protection of unvaccinated individuals.* Nature Medicine. 2021

P05 Yelin I*, **Aharony N***, Shaer-Tamar E*, Argoetti A*, Messer E, Berenbaum D, Shafran E, Kuzil A, Gandali N, Hashimshony T, Mandel-Gutfreund Y, Halberthal M, Geffen Y, Szwarcwort-Cohen M, Kishony R. *Evaluation of COVID-19 RT-qPCR test in multi-sample pools.* Clinical Infectious Diseases. 2020

P06 Moco PD, **Aharony N**, Kamen A. *Adeno-Associated Viral Vectors for Homology-Directed Generation of CAR-T cells.* Biotechnology Journal. 2019

* Equal contribution.

05 Posters and talks

Aharony N, AlQuraishi M. Evaluating Genome Language Model Embeddings as Transferable Priors for Metagenomic Binning. 2026
Accepted poster · CIBB 2026 · Rome, Italy · Sep 2–4, 2026.

Aharony N, Wang S, Neiswanger W. Benchmarking Genome Language Models for Novel Virus Detection in Wastewater Across Evolutionary Distance. 2026
Selected oral presentation · ASM BIG 2026 · Washington, DC · Oct 11–14, 2026 · Talk Oct 13.

Le Lievre T*, Aharony N*, AlQuraishi M. PRIME-Bench: A Benchmark for Pathway-Level Reasoning in Genome Language Models. 2026
Poster · ISMB 2026.

Aharony N. Who Holds the Reins in Agent-Assisted Systems Biology? 2026
Accepted poster and extended abstract · NE AI Agents Day 2026.

Aharony N, AlQuraishi M. Evaluating Genome Language Models for Metagenomic Binning. 2026
Accepted poster · Pioneering Israeli Science Conference / ScienceAbroad 2026.

Aharony N. Rapid Talks and Panel: 2024 Innovative Tools & Solutions Award Recipients. 2026
Flash talk and panelist · Pandemic Response Institute.

Aharony N. Genome Language Models Reading the Future of Pandemics in Wastewater. 2025
Flash talk · Pandemic Response Institute Innovation Showcase.

05 Posters and talks · continued

Aharony N. Computational Methods for Pandemic Pathogen Detection. 2022
Flash talk · Singer Symposium.

Aharony N, Kishony R. Rapid Alteration in Genetic Content Upon Recurring Infection. 2021
Poster · EMBL Symposium: New Approaches and Concepts in Microbiology.

Aharony N. Careers in Biosecurity. 2020
Flash talk · EAGx Virtual.

Aharony N, Gronvall G. How Secure is the Gene Synthesis Industry? 2018
Poster · Biological Weapons Convention: Meeting of Experts.

06 Awards and funding

NYC Pandemic Response Institute Innovative Tools and Solutions Award 2024
Public-health emergency preparedness research funding.

EA Funds PhD Support Scholarship 2022

Open Philanthropy Early-Career Funding 2021

Miriam and Aaron Gutwirth Memorial Fellowship for Excellence in Research 2020

Leonard and Diane Sherman Interdisciplinary Graduate School Fellowship 2019

Open Philanthropy Early-Career Funding for Global Biological Risks 2018

McGill University Faculty of Science Scholarship 2017, 2018

CIHR Undergraduate Research Award in Computational Biology 2017

NSERC Undergraduate Student Research Award · FRQNT Top-Up to NSERC Undergraduate Student Research Award 2016

07 Teaching, mentorship, and scientific service

Teaching Assistant · McGill University 2016–2018
Created problem sets, held office hours, and led review sessions for biochemistry, molecular biology, neuroscience, and introductory physics.

Research and career mentorship 2018–2023
Mentored students and early-career professionals pursuing biosecurity and computational-biology projects through Effective Thesis and 80,000 Hours referrals.

Volunteer Consultant · Battelle Memorial Institute / ThreatSEQ 2019
Advised a ThreatSEQ–FBI initiative assessing whether gene-synthesis companies would detect orders containing genes from the Select Agents list.

08 Technical and experimental methods

Python · R · MATLAB · C/C++ · Java Computational
Biological language-model evaluation · sequence modeling · metagenomic analysis · hyperbolic embeddings · genomic pipeline development

Microbial evolution experiments · AAV vector production · CRISPR/Cas9 delivery Experimental
Molecular-biology workflows

09 Languages

Hebrew
Native

English
Near-native

French
Novice